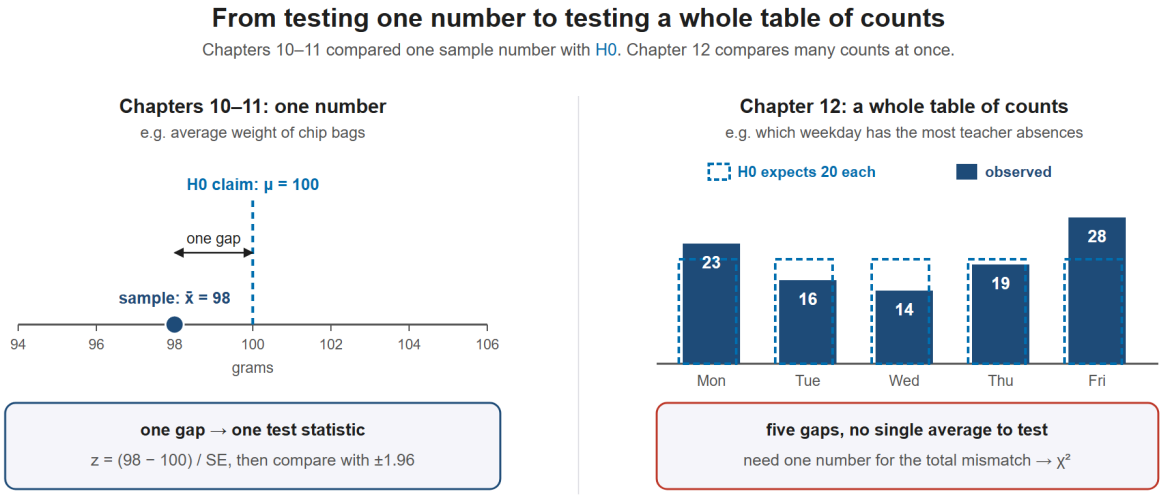
Lesson 12.1 - The Chi Square Test

> **Video:** Chi Square Test by The Organic Chemistry Tutor **Chapter:** 12 - The Chi-Square Distribution

The big picture: testing counts, not averages

Chapters 10 and 11 tested **one number**: a mean, a proportion, or a difference. Many questions are about **how counts are spread across several categories**: absences on each weekday, M&M colors in a bag, votes for 4 candidates. There is no single average to test. We need a way to compare a **whole table of counts** with what H_0 predicts.



Other count questions like this: M&M colors in a bag, votes for 4 candidates, blood types in a sample.

Left: one sample number and one claim give one gap, which becomes z or t. Right: a table of counts has a gap in every category. Chapter 12 combines all those gaps into one number, χ^2.

1. H_0 gives the expected counts. H_0 says what share each category should get. Multiply by the sample size:

$$E = n \times p$$

With 100 teachers and "absences are spread equally over 5 days", each day expects $E = 100 \times \frac{1}{5} = 20$.

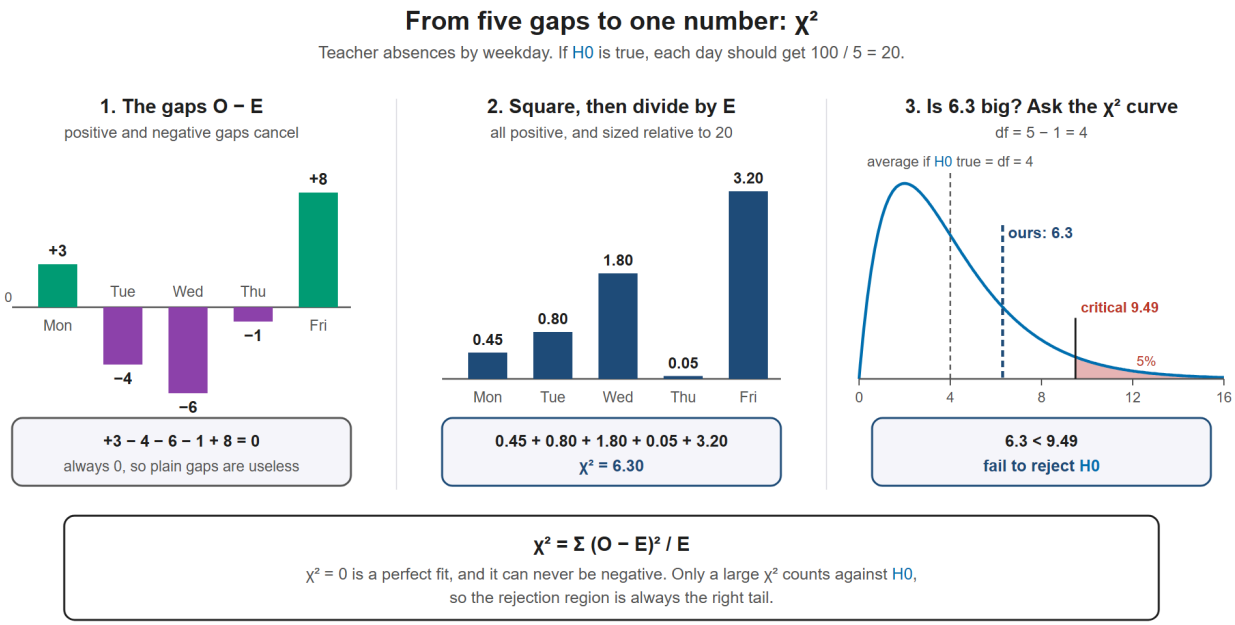
2. Why not just add up the gaps $O - E$? Because they always add up to 0. The observed and expected counts both total 100, so every extra absence on one day is a missing one somewhere else: $+3 - 4 - 6 - 1 + 8 = 0$.

3. Square, then divide by E.

- **Squaring** makes every gap positive, so the gaps can't cancel. It also gives big gaps extra weight.
- **Dividing by E** measures each gap **relative to what was expected**. Missing by 8 is a lot when you expect 20, but almost nothing when you expect 1000.

Add up the pieces for every category, and you get one number for the total mismatch:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$



The five gaps cancel out, so they are squared and scaled by E before being added. The total, 6.3, is then compared with the χ^2 curve to see whether it is unusually large.

4. Why the test is always right-tailed. χ^2 can't be negative, and $\chi^2 = 0$ means a perfect fit. Only a **large** χ^2 is evidence against H_0, so the rejection region is always the right tail. A χ^2 near 0 just means the data fit very well.

5. Where the χ^2 curve comes from. Each piece $\frac{O-E}{\sqrt{E}}$ behaves roughly like a z-score, so χ^2 is roughly a **sum of squared z-scores**. That sum has its own distribution, the **chi-square distribution**:

- It starts at 0 and is skewed right, because squares are never negative.
- Its shape depends on the **degrees of freedom**. On average, χ^2 is about **df** when H_0 is true. Here df = 4 and we got 6.3, a bit more than average but not unusual.

The chi-square distribution formula. If $Z_1, \ldots, Z_k$ are independent standard normal variables, then

$$\chi^2 = Z_1^2 + Z_2^2 + \cdots + Z_k^2$$

follows a chi-square distribution with k degrees of freedom. Its curve (probability density function) is

$$f(x) = \frac{x^{k/2-1}\, e^{-x/2}}{2^{k/2}\, \Gamma\left(\frac{k}{2}\right)}, \qquad x > 0$$

Quantity	Formula
Mean	$\mu = k$
Variance	$\sigma^2 = 2k$
Standard deviation	$\sigma = \sqrt{2k}$

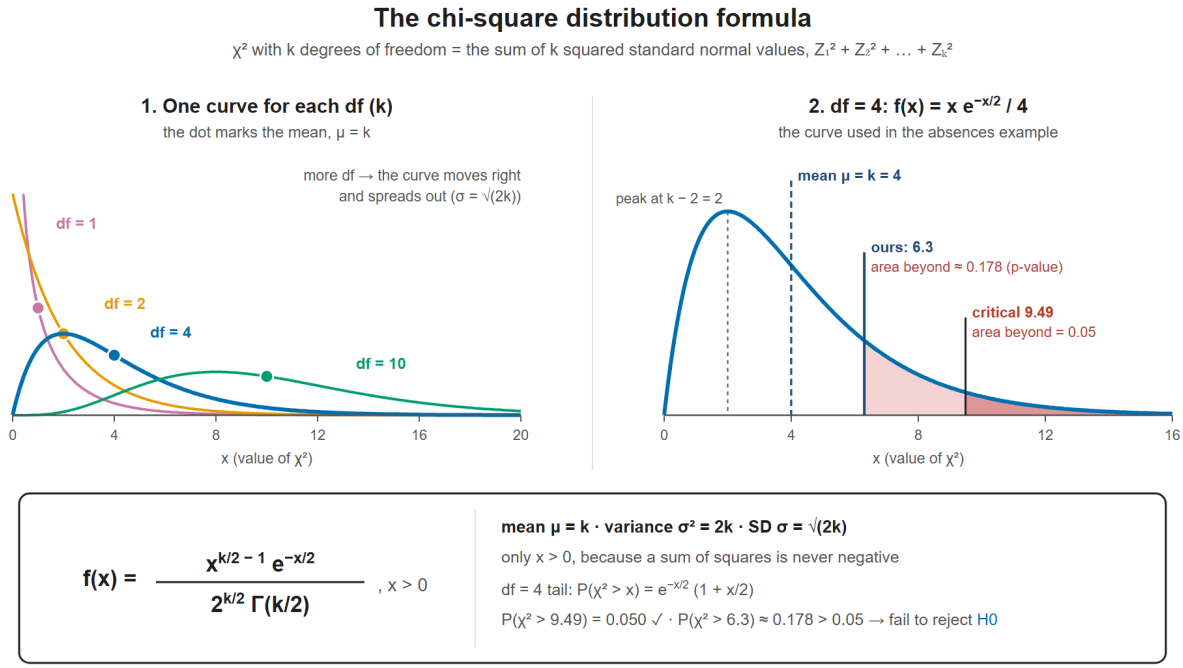
- k is the degrees of freedom (df).
- Γ is the gamma function, which extends the factorial: $\Gamma(n) = (n-1)!$ for whole numbers, and $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.
- $x > 0$ only, because a sum of squares can't be negative. That is why the curve starts at 0.
- The mean $\mu = k$ is why χ^2 "averages about df" when H_0 is true.

Example: df = 4. Here $\Gamma(2) = 1! = 1$ and $2^{4/2} = 4$, so the curve is

$$f(x) = \frac{x^{\,4/2-1}\, e^{-x/2}}{2^{4/2}\,\Gamma(2)} = \frac{x\, e^{-x/2}}{4}$$

For df = 4 the right-tail area also has a short closed form, $P(\chi^2 > x) = e^{-x/2}\left(1 + \frac{x}{2}\right)$. This checks the table and gives the p-value:

- At the critical value: $P(\chi^2 > 9.49) = e^{-4.745}(1 + 4.745) \approx 0.0087 \times 5.745 \approx 0.050$ ✓. That is exactly the 5% the table promises.
- At our value: $P(\chi^2 > 6.3) = e^{-3.15}(1 + 3.15) \approx 0.0429 \times 4.15 \approx 0.178$. The p-value is about 0.18, which is greater than 0.05, so we fail to reject H_0.



Left: each df has its own curve. The mean sits at k, so the curve slides right and spreads out as df grows. Right: the df = 4 curve from the absences example. The p-value (light red, about 0.178) is the area beyond our 6.3. The darker tail beyond 9.49 is the 5% rejection region.

You won't usually compute $f(x)$ by hand. The χ^2 table (or software) already lists the areas. The formula explains the curve's shape: it starts at 0, rises, then has a long right tail.

6. Why df $= k - 1$. The counts must add up to the total. Once you know 4 of the 5 days, the 5th is forced: $100 - 23 - 16 - 14 - 19 = 28$. Only $k - 1$ counts are free to vary.

The 5 steps are the same as in Chapters 10 and 11:

1. Write H_0 (the expected distribution) and H_a (it doesn't fit).
2. Choose α.
3. Find the critical value from the χ^2 table, with df $= k - 1$ and right-tail area α.
4. **Compute** $\chi^2 = \sum(O - E)^2/E$. ← this lesson
5. χ^2 greater than the critical value → reject H_0. Otherwise → fail to reject H_0.

The goodness-of-fit test

Does the **observed** distribution of counts across categories match an **expected** distribution?

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

- O = observed count, E = expected count, for each category
- **Degrees of freedom** = $k-1$, where k = number of categories
- It is always a **right-tailed** test: a large χ^2 means the observed counts are far from the expected ones

The chi-square distribution is **not symmetric**; it is skewed to the right. The critical value χ^2_α cuts off area α in the right tail:

- calculated χ^2 > critical value → **rejection region** → reject H_0
- calculated χ^2 ≤ critical value → **do-not-reject region**

Example - which weekday has the most absences?

A principal expects absences to be spread **equally** over the 5-day school week. 100 teachers were asked which day had the most absences. Test at $\alpha = 0.05$.

Day	Mon	Tue	Wed	Thu	Fri	Total
Observed O	23	16	14	19	28	100
Expected E	20	20	20	20	20	100

Expected counts: $E = n \times p = 100 \times \frac{1}{5} = 20$ for each day.

Hypotheses:

- H_0: absences happen with **equal frequencies** (a uniform distribution)
- H_a: absences do **not** happen with equal frequencies

Critical value: df $= k-1 = 5-1 = 4$. In the χ² table with right-tail area 0.05: $\chi^2_{0.05,\,4} = 9.49$.

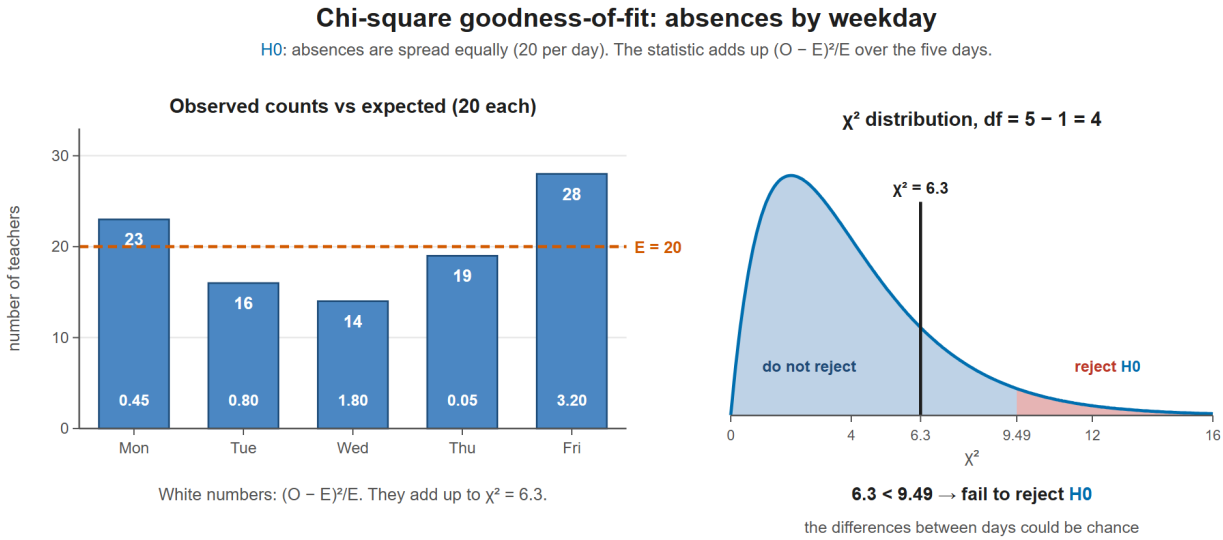
Test statistic:

Day	$O-E$	$(O-E)^2$	$(O-E)^2/E$
Mon	3	9	0.45
Tue	−4	16	0.80
Wed	−6	36	1.80
Thu	−1	1	0.05
Fri	8	64	3.20
Sum		**126**	**6.30**

$$\chi^2 = \sum \frac{(O-E)^2}{E} = \frac{9+16+36+1+64}{20} = \frac{126}{20} = 6.3$$

(Every E is 20 here, so you can add the squares first and divide once.)

$6.3 < 9.49$, which is in the do-not-reject region. **Fail to reject H_0.** The variation between days is not large enough to say that absences are unevenly spread across the week.



Left: Friday and Wednesday are furthest from the expected 20, so they contribute the most to χ^2. Right: the total, 6.3, is still below the critical value 9.49, so the uneven pattern could be due to chance.

Key takeaways

- The gaps $O - E$ always add up to 0, so they are squared and divided by E before being added.
- $\chi^2 = \sum (O - E)^2 / E$ measures how far the observed counts are from the expected ones.
- Goodness of fit: df = number of categories − 1. It is always right-tailed.
- Large χ^2 (beyond the critical value) → reject H_0 (the data don't fit the expected distribution).